

SIVA BALAN G

AI/ML Engineer

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PROFESSIONAL SUMMARY

Motivated AI and Machine Learning Engineer (Fresher) with hands-on experience building intelligent systems using Machine Learning and Large Language Models (LLMs). Skilled in Python, Supervised and Unsupervised, Embedding, Retrieval-Augmented Generation (RAG), and Vector Databases. Experienced in developing end-to-end ML pipelines, model deployment, and scalable AI applications.

TECHNICAL SKILLS

Programming:	Python,SQL
DataScience:	NumPy, Pandas, Scikit-learn
Machine Learning:	Supervised & Unsupervised Learning,Feature Engineering,Model Evaluation

Generative AI:	LLMs, LangChain, RAG, Prompt Engineering, Hugging Face
Vector Databases:	ChromaDB, FAISS
Deployment and Tools:	Fast API, REST APIs, Git, Jupyter Notebook,VS Code

PROJECTS

Document QA Intelligent Systems

- Built an intelligent multimodal RAG system for context-aware Q/A across text, images, and layout-rich documents
- Designed end-to-end document ingestion workflow with extraction, intelligent chunking, and metadata enrichment
- Implemented semantic search using vector embeddings to surface relevant content segments per query
- Integrated LLM to generate precise, source-grounded responses with full traceability

Object Detection using YOLO

- Developed a computer vision-based object detection system using YOLO model to detect manufacturing defects in metal plates, including misaligned holes and incorrectly positioned components.
- Curated and annotated defect datasets and configured the detection model to accurately localize structural and alignment-related faults
- Optimized the detection pipeline for real-time inference to enhance quality control, reduce manual inspection, and improve manufacturing efficiency.

Customer Lifetime Value Prediction

- Implemented a Customer Lifetime Value prediction project using Python, Pandas, and XGBoost, including data preprocessing, feature selection, and model training.
- Evaluated model performance and handled data inconsistencies such as outliers and missing values.

EDUCATION

2021 - 2025	B.Tech AI&DS at SIET	[CGPA:7.21]
2020 - 2021	HSC	[91.5%]
2018 - 2019	SSLC	[93%]

CERTIFICATION

AI and ML, Gen AI - Proitbridge Software Training Institute